A New Perspective on Target-Conditioned Structural Dynamics for Link Prediction in Dynamic Graphs

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Abstract

Dynamic link prediction requires determining whether the historical interactions of a source node provide reliable structural support for a specific target. Existing target-conditioned methods capture such support through structural heuristics such as repeated interactions or co-neighbor overlap, but typically instantiate them as identity-based encodings and inject them as passive token-level features before aggregation. This design is brittle when exact structural encodings are sparse, and the resulting structural signals can be diluted after being concatenated with high-dimensional time and edge features, while their dynamics remain unmodeled. In this paper, we revisit target-conditioned dynamic link prediction from a structural dynamics modeling perspective and propose TCSD, which formulates structural signals as target-conditioned states, evolves them over recent histories, and uses them to guide aggregation. Concretely, we construct hybrid structural states from exact indicators that preserve time-aware repeat and co-neighbor matches, and relaxed indicators that recover latent support beyond exact identity matching. TCSD further models the dynamics of these states to capture temporal consistency within indicators and complementarity across indicators. Instead of treating structural states as passive features, TCSD uses the learned dynamics as aggregation controllers to amplify target-relevant interactions and suppress irrelevant ones. Experiments on sixteen dynamic graphs show that TCSD outperforms ten baselines, achieving up to a 14.27% relative improvement in MRR. The code is available at https://github.com/LuckyGirl-XU/TCSD

1 Introduction

Dynamic link prediction is a fundamental task for modeling evolving graphs, with broad applications in recommendation systems [1, 2, 3, 4, 5], social network analysis [6, 7, 8, 9], and financial prediction [10, 11, 12]. Given a source node and a candidate target at query time, the goal is to determine whether the source's historical interactions provide reliable support for the prediction. This makes target conditioning essential: the same source history may be highly informative for one target but irrelevant for another. Existing target-conditioned methods [13, 14, 15, 16, 17, 18, 19, 20, 21, 22] mainly exploit structural heuristics such as repeated interactions [16, 22, 18] or co-neighbor overlap [19, 20, 21], where the candidate target is compared with the source historical interactions or

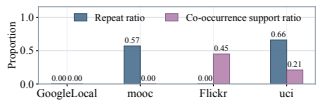
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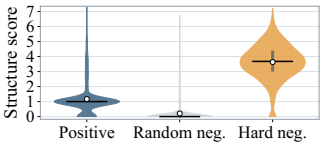
its temporal neighborhood. These methods have shown the effectiveness of target-specific structure modeling, but they typically instantiate such signals as static identity-based encodings and inject them as passive token-level features before aggregation. This design has two key limitations.

Limitation I: Existing exact identity-based structural encodings are brittle across diverse dynamic graphs. As shown in Fig. 1a, different datasets exhibit substantially different structural regimes. Mooc is mainly repeat-driven (57%), uci contains both repeat (66%) and co-neighbor support (21%), while GoogleLocal provides almost no exact support of either type (0%). In such low-support regimes, existing structural encodings, such as co-neighbor encodings in DyGFormer [19] and SALoM [21], as well as repeat encodings in CRAFT [22], may collapse to uninformative zero vectors, even though latent target-relevant structure may still exist. This suggests that structural support should not rely solely on exact identity matching; instead, it should be represented through both exact and relaxed structural indicators that preserve high-confidence matches when available while recovering latent support beyond strict identity matching.

Limitation II: Static feature concatenation dilutes and may mislead structural signals. Even when target-conditioned structural support is available, existing methods [19, 20, 21, 22, 17] typically concatenate structural encodings with high-dimensional time and edge features before aggregation. This design reduces structural cues to passive token-level features, whose signals may be diluted during feature mixing. Moreover, static structural encodings do not model how such support evolves over recent source histories and may therefore over-emphasize isolated or outdated matches. This issue becomes more pronounced under hard negatives. As shown in Fig. 1b, identity-based statistics computed from repeat interactions and co-neighbor overlap within 20 recent neighbors assign higher structural scores to hard negatives (around 3.68) than to positive targets (around 1.00), indicating that exact matches alone can produce structurally plausible but incorrect support when its temporal dynamics are ignored. Therefore, reliable dynamic link prediction requires modeling the temporal consistency and dynamics of target-conditioned structural support and using them to guide aggregation.



(a) Overall structure statistics



(b) Identity-based statistics

Figure 1: Structural statistics across datasets and identity-based scores on uci, where hard negatives are detailed in Section 5.4.

To address these limitations, we revisit dynamic link prediction from a structural dynamics modeling perspective. We propose TCSD, a **T**arget-**C**onditioned **S**tructural **D**ynamics framework for link prediction in dynamic graphs. The key idea is to transform structural signals from passive token-level features into dynamic guidance for aggregation. Concretely, we first design hybrid structural indicators that convert historical interactions into target-conditioned structural states, including exact indicators for time-aware repeat and co-neighbor matches, and relaxed indicators that recover latent structural support beyond exact identity matching, thereby alleviating encoding collapse in sparse settings and addressing **Limitation I**. TCSD then models the dynamics of these states with temporal convolutional module to capture temporal consistency within an indicator and complementarity across indicators. Instead of injecting the learned structural dynamics as passive features (**Limitation II**), TCSD converts them into structural gates that control each neighbor's contribution during aggregation. Thus, TCSD explicitly models structural dynamics and uses them to guide target-conditioned aggregation for more reliable link prediction.

Our contributions are summarized as:

- We propose a new structural dynamics modeling perspective for dynamic link prediction, which shifts structural signals from static passive feature injection to dynamic state modeling and active aggregation guidance.
- We instantiate this perspective with TCSD, which constructs target-conditioned structural states via hybrid structural indicators as the basis, models their dynamics over recent histories to capture temporal consistency and complementarity across indicators, and translates the learned dynamics into structural gates for neighbor aggregation control.
- Extensive experiments on sixteen dynamic graphs show that TCSD outperforms ten baselines, achieving up to a 14.27% MRR improvement. Further analyses show that hybrid structural states and dynamics-guided aggregation improve hard-negative disambiguation.

2 Related Work

General dynamic graph modeling. General dynamic graph modeling methods primarily focus on exploring node-level temporal and structural dependencies for link prediction. Existing approaches can be broadly categorized into sequence-based methods [23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33], memory-based methods [34, 35, 21, 36, 37, 38, 39, 40], classical message passing methods [41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51], frequency-domain modeling methods [52, 53, 54], and hyperbolic geometry-based methods [55, 56]. Despite their effectiveness, most of them learn general-purpose node representations that are not explicitly conditioned on a specific candidate link, leading to suboptimal performance.

Target-conditioned dynamic graph modeling. Target-conditioned modeling on dynamic graphs learns representations conditioned on a target, which adaptively constructs temporal and structural signals for scoring a specific link. Existing methods typically introduce target conditioning through two types of structural heuristics. The first line of work [13, 14, 15, 17, 20, 19, 21, 57] relies on co-neighbor overlap, where the candidate source and destination are compared through their shared neighborhoods, e.g., DyGFormer [19] and SALoM [21]. The second line [16, 18, 22] focuses on repeat patterns, i.e., whether the candidate destination has appeared in the source node's history before, as in RepeatMixer [16] and CRAFT [22]. These methods show the significance of target-conditioned structural cues. However, they are often designed around a single type of cue: co-neighbor-based methods emphasize local topological overlap, while repeat-based methods emphasize pairwise recurrence. Consequently, they can be less robust when the corresponding structural patterns are sparse or absent. Moreover, these structural encodings are usually treated as token features and mixed with edge and time features before aggregation, leaving their structural dynamics underexplored.

3 Preliminaries

Continuous-Time Dynamic Graph. We consider a continuous-time dynamic graph (CTDG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the node set and $\mathcal{E}$ is the set of temporal interactions. Each interaction is represented as $(u, v, t) \in \mathcal{E}$, with $u, v \in \mathcal{V}$ denoting the source and destination nodes and t denoting the interaction timestamp. When edge attributes are available, we write an interaction as $(u, v, t, \mathbf{e}_{uv}(t))$, where $\mathbf{e}_{uv}(t)$ is the edge feature.

Problem Definition. Given a query (s, d, t) with a source node s, a candidate destination d, and a timestamp t, the goal of dynamic link prediction is to estimate the probability that s will interact with d at time t conditioned on the historical interactions before t.

4 Methodology

In this paper, we propose a target-conditioned structural dynamics (TCSD) framework for link prediction in dynamic graphs. As shown in Fig. 2, TCSD organizes temporal structure modeling into three stages: hybrid structural state construction, structural dynamics modeling, and aggregation controller. Instead of treating structural signals as static passive token features for aggregation, TCSD first constructs target-conditioned structural states as the basis, then models these states to capture structural dynamics, and finally uses the learned dynamics as aggregation controller for link scoring. We detail each module below.

4.1 Target-Conditioned Structural State Construction

A central challenge in dynamic graph learning is extracting reliable structural support from historical interactions, for which target-conditioned modeling has proven effective. Recent methods [19, 20, 21, 16, 22, 18] typically construct neighbor encodings using exact structural heuristics, such as repeated interactions or co-neighbor overlap. While effective when explicit matches exist, these heuristics are brittle under sparse structural regimes, as discussed in Limitation I and shown in Fig. 8 of Appendix B. To construct more robust structural states, we introduce hybrid structural indicators that capture repeat- and co-neighbor-based support through both exact identity matching and relaxed embedding affinity. Given a query (s, d, t), where s is the source node, d is the candidate destination, and t is

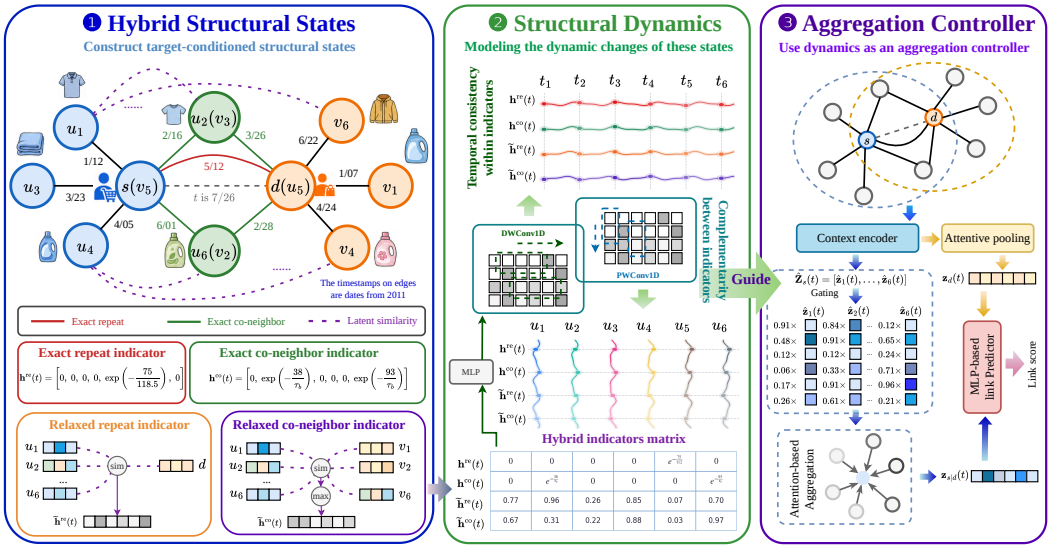


Figure 2: Overview of our target-conditioned structural dynamics framework, including structural state construction, state dynamic modeling, and dynamics-guided aggregation for link prediction.

the timestamp, let $\mathcal{N}_s(t) = \{(u_i, t_i)\}_{i=1}^{L}$ denote the L recent neighbors of source node s before t sampled by recent sampling [19]. Similarly, let $\mathcal{N}_d(t) = \{(v_j, t_j)\}_{j=1}^{L}$ denote the L recent neighbors of candidate destination d before t. We present hybrid indicators to construct structural states below.

Exact structural indicators. Existing exact neighbor encodings [19, 20, 21, 16, 22, 18] mainly indicate whether repeated interactions or co-neighbor overlaps exist, but often ignore their temporal recency. As a result, they may assign similar support to recent positive evidence and outdated hard-negative ones. To preserve exact structural cues while accounting for temporal alignment, we define time-aware exact indicators: exact repeat and co-neighbor indicators. For repeated interactions, since a previously observed query pair provides high-confidence support, we define the time-aware exact repeat indicator for each source-side neighbor u_i conditioned on destination d at time t as

$$h_i^{\mathrm{re}}(t) = \mathbb{I}(u_i = d) \exp\left(-\frac{t - t_i}{\tau_s}\right), \ (u_i, t_i) \in \mathcal{N}_s(t), \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function, $\tau_s = \mathrm{median}(\{t - t_i\}_{(u_i,t_i)\in\mathcal{N}_s(t)})$ denotes the median time gap, and $\exp(\cdot)$ denotes the exponential function. This indicator activates when $u_i = d$, with exponential decay assigning larger weights to more recent repetitions, yielding a time-aware exact repeat signal.

Co-neighbor overlap captures whether the source and destination share historical neighbors. For each source-side neighbor $(u_i, t_i) \in \mathcal{N}_s(t)$, we define the matched destination-side index set as $\mathcal{J}_i(t) = \{j \mid (v_j, t_j) \in \mathcal{N}_d(t),\ v_j = u_i\}$. The time-aware exact co-neighbor indicator is defined as

$$h_i^{\mathrm{co}}(t) = \mathbb{I}(\mathcal{J}_i(t) \neq \emptyset) \exp\left(-\frac{\min\limits_{j\in\mathcal{J}_i(t)} |t_i - t_j|}{\tau_b}\right), \tag{2}$$

where $\tau_b > 0$ is a learnable time scale. If $\mathcal{J}_i(t) = \emptyset$, we directly set $h_i^{\mathrm{co}}(t) = 0$ and the minimum term is not evaluated. Otherwise, the indicator assigns higher scores to temporally aligned co-neighbor matches and down-weights temporally misaligned overlaps.

Relaxed structural indicators. Similar to existing identity-based statistics, our exact indicators may collapse under sparse observations because they rely on exact identity matches. Fig. 2 illustrates this issue. Since $u_5 = d$ and only u_2 and u_6 overlap with $\mathcal{N}_d(t)$, nonzero strengths are assigned only to these exact-match positions: $\mathbf{h}^{\mathrm{re}}(t) = [0, 0, 0, 0, \exp(-75/118.5), 0]$ and $\mathbf{h}^{\mathrm{co}}(t) = [0, \exp(-38/\tau_b), 0, 0, 0, \exp(-93/\tau_b)]$. However, zero-valued positions may still contain latent structural relevance, e.g., u_4 and v_4 correspond to laundry detergents from different

brands. To address this, we introduce two relaxed structural indicators based on embedding similarity: relaxed repeat and co-neighbor indicators. Given the embedding $\mathbf{x}_i(t_i)$ of each source-side neighbor $(u_i, t_i) \in \mathcal{N}_s(t)$, the embedding $\mathbf{x}_d(t)$ of the candidate destination d, and the embeddings $\{\mathbf{x}_j(t_j)\}_{v_j \in \mathcal{N}_d(t)}$ of destination-side historical neighbors, we define relaxed repeat indicator $\widetilde{h}_i^{\mathrm{re}}(t)$ and relaxed co-neighbor indicator $\widetilde{h}_i^{\mathrm{co}}(t)$ as

$$\widetilde{h}_i^{\mathrm{re}}(t) = \mathrm{sim}\left(\mathbf{x}_i(t_i), \mathbf{x}_d(t)\right), \ \widetilde{h}_i^{\mathrm{co}}(t) = \max_{1 \le j \le L} \mathrm{sim}\left(\mathbf{x}_i(t_i), \mathbf{x}_j(t_j)\right), \tag{3}$$

where $\mathrm{sim}(\cdot, \cdot)$ denotes cosine similarity. In our implementation, $\mathbf{x}_*(\cdot)$ is initialized from node features and is updated only before the corresponding timestamp, avoiding temporal leakage. The relaxed repeat indicator measures latent compatibility between a source's neighbor and the candidate destination, while the relaxed co-neighbor indicator captures the strongest embedding-level affinity between the source-side neighbor and the destination-side history. Unlike exact indicators, relaxed indicators do not require identical node matches and therefore remain informative when the candidate destination is absent from the source history or when the two neighborhoods have no exact overlap.

Hybrid structural states. Exact indicators preserve high-confidence structural matches in dense-support regimes, while relaxed indicators provide latent compatibility. For each source-side neighbor, we concatenate these indicators to obtain target-conditioned structural states:

$$\mathbf{h}_i(t) = \left[h_i^{\mathrm{re}}(t) \,\middle\|\, h_i^{\mathrm{co}}(t) \,\middle\|\, \widetilde{h}_i^{\mathrm{re}}(t) \,\middle\|\, \widetilde{h}_i^{\mathrm{co}}(t) \right], \tag{4}$$

where $\|$ denotes concatenation. The state $\mathbf{h}_i(t)$ summarizes target-conditioned structural states from both exact and relaxed repeat/co-neighbor signals. This hybrid design preserves high-confidence structural evidence when available, while maintaining informative support in sparse regimes, providing robust basis for aggregation.

To verify the necessity of hybrid structural indicators, we measure the separability of each indicator c by the gap $\Delta_c = \mathbb{E}[\bar{H}_c^+] - \mathbb{E}[\bar{H}_c^-]$, where $\bar{H}_c^+$ and $\bar{H}_c^-$ denote the average indicator scores over positive and negative targets, respectively; see Appendix C.3 for the formal definition and analysis. As shown in Fig. 3, different indicators provide complementary discriminative signals across graph regimes, and relaxed indicators yield more robust structural separability.

Insight. Existing identity-based statistics provide high-confidence evidence when exact matches are available, but their effectiveness deteriorates under sparse structural support. Hybrid structural indicators improve robustness by combining exact channels with relaxed embedding-based channels, allowing TCSD to exploit explicit structural patterns in dense regimes while preserving informative support in unseen or low-support regimes.

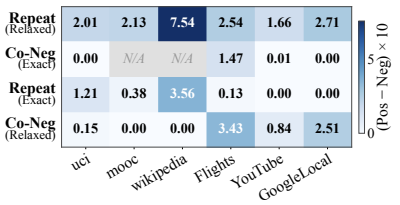


Figure 3: Complementarity of hybrid indicators across six datasets.

4.2 Structural Dynamics-Guided Aggregation

As discussed in Limitation II, simply concatenating structural encodings with edge and time features reduces structural signals to passive token-level attributes, which obscures their contribution and limits the effectiveness of neighbor encoding. Given the hybrid structural states constructed above, we instead model how these states evolve over recent source histories and use the learned dynamics to guide aggregation. This changes the role of structural information from static feature injection to active aggregation control: temporally consistent target-relevant support can be emphasized, while isolated or outdated signals can be suppressed. We first encode source and destination contexts without injecting hybrid structural states, and then introduce structural dynamics-guided aggregation.

Encoding source and destination contexts. We first encode the temporal contexts of the source and candidate destination for a query (s, d, t). For each source-side historical interaction $(u_i, t_i) \in \mathcal{N}_s(t)$, we construct the input token by combining its neighbor embeddings, time encoding, and edge feature:

$\bar{\mathbf{x}}_i(t) = \phi_x([\mathbf{x}_i(t^-) \,\|\, \phi_t(t - t_i) \,\|\, \mathbf{e}_{s,i}(t_i)])$, where t^- denotes the last time and $\phi_x(\cdot)$ is the linear projection. We then apply an attention-based encoder $\Phi(\cdot)$ [58] to the source-history sequence:

$$\hat{\mathbf{Z}}_s(t) = \Phi\big(\bar{\mathbf{X}}_s(t)\big)\,, \text{ where } \bar{\mathbf{X}}_s(t) = [\bar{\mathbf{x}}_1(t), \ldots, \bar{\mathbf{x}}_L(t)]. \tag{5}$$

The output $\hat{\mathbf{Z}}_s(t) = [\hat{\mathbf{z}}_1(t), \ldots, \hat{\mathbf{z}}_L(t)]$ preserves the contextualized source-history representations. Similarly, we encode the destination-side history $\mathcal{N}_d(t)$ and obtain the contextualized destination-history representations $\hat{\mathbf{Z}}_d(t)$. Then, attentive pooling is applied to $\hat{\mathbf{Z}}_d(t)$ to derive the candidate destination representation $\mathbf{z}_d(t)$. These context representations are later combined with structural dynamics for target-conditioned aggregation.

Modeling structural dynamics. Given the hybrid structural states, we obtain a target-conditioned state sequence $\mathbf{H}(t) = [\mathbf{h}_1(t), \ldots, \mathbf{h}_L(t)]$ over the source-side history. This sequence should not be collapsed into a static summary: two candidates may have similar overlap counts, while one receives support from several recent positions and the other only from an outdated interaction. To capture such temporal patterns, we model local structural dynamics along the recent interactions by

$$\mathbf{R}(t) = \mathrm{MLP}\left(\mathrm{PWConv1D}\left(\mathrm{DWConv1D}\left(\mathbf{H}(t)\right)\right)\right). \tag{6}$$

Here, $\mathrm{DWConv1D}(\cdot)$ performs depthwise convolution along the source-history dimension, capturing local temporal consistency within each structural channel, such as whether repeat or co-neighbor support is persistent or isolated. $\mathrm{PWConv1D}(\cdot)$ then mixes structural channels, allowing different indicators to reinforce or calibrate each other. $\mathrm{MLP}(\cdot)$ denotes a multi-layer perceptron. The convolutions in Eq. (6) adaptively calibrate the heterogeneous scales and signs of raw hybrid indicator channels before gating. The output $\mathbf{R}(t) = [\mathbf{r}_1(t), \ldots, \mathbf{r}_L(t)]$ is the evolved structural state sequence.

Using structural dynamics as an aggregation controller. The evolved structural states indicate which source-side neighbors provide temporally consistent support for the candidate target. However, directly concatenating them to neighbor tokens would still treat structural information as passive features. Instead, we convert the structural state and its dynamics into a dimension-wise structural gate to modulate neighbor representations and formulate it as

$$\widetilde{\mathbf{z}}_i(t) = \mathbf{g}_i(t) \odot \hat{\mathbf{z}}_i(t),\ \mathbf{g}_i(t) = \sigma\left(\mathrm{MLP}\left([\mathbf{h}_i(t) \,\|\, \mathbf{r}_i(t)]\right)\right), \tag{7}$$

where $\sigma(\cdot)$ is the sigmoid function, and $\odot$ denotes element-wise multiplication. The gate $\mathbf{g}_i(t)$ serves as a dimension-wise controller for the contextualized representation of the i-th historical neighbor under candidate target d. We aggregate the gated neighbor representations into a target-conditioned source representation:

$$\mathbf{z}_{s|d}(t) = \sum_{i=1}^{L} \alpha_i(t)\, \widetilde{\mathbf{z}}_i(t), \quad \alpha_i(t) = \frac{\exp\left(f_a(\widetilde{\mathbf{z}}_i(t))\right)}{\sum_{k=1}^{L} \exp\left(f_a(\widetilde{\mathbf{z}}_k(t))\right)}, \tag{8}$$

where $f_a(\cdot)$ is an MLP that maps each gated representation to a scalar logit, $\alpha_i(t)$ is the softmax-normalized pooling weight, and $\mathbf{z}_{s|d}(t)$ is the source representation conditioned on d. The structural gate $\mathbf{g}_i(t)$ in Eq. (7) suppresses irrelevant dimensions and preserves target-relevant ones. The subsequent score-based pooling of Eq. (8) assigns larger weights to neighbors whose gated representations contain stronger target-relevant structural support, as shown in Fig. 2③. Finally, an MLP predictor scores (s, d, t) from $\mathbf{z}_{s|d}(t)$ and $\mathbf{z}_d(t)$. Overall, structural dynamics-guided aggregation turns structural information from passive features into active aggregation control, helping distinguish temporally consistent support from spurious static overlap.

> **Insight.** Dynamic link prediction depends not only on the existence of target-conditioned structural support but also on its temporal consistency over recent histories. Static identity-based statistics ignore these temporal dynamics and may assign high scores to hard negatives due to isolated or outdated matches. Therefore, structural signals should be modeled as evolving states and used to regulate aggregation, emphasizing consistent target-relevant evidence while suppressing spurious matches for link prediction.

Table 1: MRR scores (%) of TCSD and nine baseline methods on unseen-dominant datasets. The **best** and second-best results are highlighted accordingly. "-" denotes TLE or OOM in our environment. $*$ indicates results collected from [22].

Datasets	ML-20M	Taobao	GoogleLocal	Flickr	YouTube	WikiLink	tgbl-review	tgbl-comment
JODIE*	21.16±0.73	48.36±2.18	41.86±1.49	46.21±0.83	41.67±2.86	57.94±1.33	41.43±0.15	-
DyRep*	19.00±1.69	40.03±2.40	37.73±1.34	38.04±4.19	35.12±4.13	42.63±1.33	40.06±0.59	28.90±3.30
TGAT*	10.47±0.20	-	19.78±0.24	23.53±3.35	43.56±2.53	-	19.64±0.23	56.20±2.11
TGN*	23.99±0.20	60.28±0.54	54.13±1.97	46.03±6.78	55.16±5.89	62.94±2.16	37.48±0.23	37.90±2.10
GraphMixer*	21.97±0.17	31.54±0.02	21.31±0.14	45.01±0.08	58.87±0.12	48.57±0.02	36.89±1.50	76.17±0.17
CAWN*	12.31±0.02	-	18.26±0.02	48.69±6.08	47.55±1.08	-	19.30±0.10	-
DyGFormer*	-	-	18.39±0.02	49.58±2.87	46.08±3.44	-	22.39±1.52	67.03±0.14
CRAFT*	**35.91±0.65**	70.68±0.42	62.35±0.46	62.34±0.84	58.92±0.16	75.48±0.84	41.77±0.06	91.72±0.59
TAMI-GM	33.86±0.57	42.70±0.55	22.87±0.63	48.69±0.78	45.71±0.44	-	44.64±0.19	-
TCSD	35.10±0.58	**73.34±0.46**	**66.24±0.55**	**69.99±0.00**	**67.33±0.35**	**79.23±0.14**	**46.09±0.39**	**92.73±0.09**
Rel. Imprv.	-2.26%	+3.76%	+6.24%	+12.27%	+14.27%	+4.97%	+3.25%	+1.10%

Table 2: MRR scores (%) of TCSD and ten baseline methods on seen-dominant datasets.

Datasets	wikipedia	reddit	mooc	lastfm	uci	Flights	tgbl-coin	tgbl-flights
JODIE*	76.48±1.72	77.16±1.27	19.91±3.06	18.49±2.87	51.43±1.74	20.37±4.18	-	-
DyRep*	67.42±4.21	72.33±2.14	17.71±1.27	17.76±6.56	14.00±2.71	15.62±0.47	45.20±4.60	55.60±1.40
TGAT*	72.72±1.50	78.03±0.25	31.55±4.14	24.10±1.26	31.50±0.48	53.90±1.07	60.92±0.57	-
TGN*	82.22±0.39	79.25±0.24	39.03±4.53	25.59±4.43	45.29±6.43	64.24±2.11	58.60±3.70	70.50±2.00
GraphMixer*	73.57±1.48	70.87±0.38	29.02±0.28	26.76±1.42	59.58±1.00	42.26±0.49	75.57±0.27	-
CAWN*	86.07±0.08	87.66±0.04	27.02±0.57	35.10±0.34	66.96±0.48	80.06±2.10	-	-
DyGFormer*	88.69±0.04	88.70±0.05	42.20±1.31	46.50±0.51	76.61±0.26	84.13±3.46	75.17±0.38	-
CRAFT*	88.25±0.26	89.33±0.11	62.32±0.39	55.21±0.23	75.11±0.09	83.16±0.16	88.47±0.25	**91.39±0.03**
TAMI-GM	88.47±0.05	86.38±0.04	29.23±0.10	38.86±0.08	68.12±0.12	75.53±0.09	-	-
TAMI-DyG	86.99±0.04	88.39±0.01	35.11±0.25	45.30±0.18	74.65±0.17	**85.30±0.51**	-	-
TCSD	**89.51±0.02**	**89.42±0.14**	**69.78±0.38**	**55.88±0.45**	**79.50±0.19**	83.14±0.51	**91.12±0.22**	87.58±0.37
Rel. Imprv.	+0.92%	+0.10%	+11.97%	+1.21%	+3.77%	-2.53%	+3.00%	-4.17%

Theoretical statement. For fixed contextualized neighbor representations, the aggregation-stage passive structural feature injection family with identity or affine value projection before pooling is contained in the uniform closure of the TCSD aggregation with constant auxiliary coordinates. The formal proposition and proof are given in the Appendix A.1.

5 Experiments

5.1 Experimental Settings

Datasets, Baselines, and Evaluation protocol. We evaluate TCSD on 16 dynamic graph datasets covering both seen-dominant and unseen-dominant settings [22, 59]. We compare TCSD with ten baselines, including general dynamic graph models (JODIE [23], DyRep [41], TGAT [42], TGN [34], and GraphMixer [43]) and target-conditioned methods based on co-neighbor or repeat evidence (CAWN [13], DyGFormer [19], CRAFT [22], and TAMI [18] with GraphMixer and DyGFormer). We follow the ranking-based future link prediction setting and use MRR as the main evaluation metric. For each positive query, we randomly sample 100 negative destinations. We report the mean and standard deviation over 3 runs. TLE denotes exceeding 48 hours per run on one V100 GPU; OOM denotes exceeding 32 GB of GPU memory under the same batch size. More experimental details are in the Appendix C.1.

5.2 Effectiveness Evaluation

We compare TCSD with ten competitive baselines on both seen-dominant and unseen-dominant dynamic graphs, and report the MRR results in Tables 1 and 2.

EXP-1: Effectiveness of the proposed TCSD for unseen prediction. TCSD achieves an average relative MRR improvement of 5.45% across eight unseen-dominant datasets, with substantial gains

on GoogleLocal, Flickr, and YouTube of 6.24%, 12.27%, and 14.27%, respectively. These results highlight the advantage of TCSD on dynamic graphs dominated by new interactions, where repeated interactions are rare (see Table 5). The gains stem from TCSD's ability to construct structural states, model their dynamics, and use the learned dynamics to guide aggregation. In contrast, general dynamic graph models lack target-specific structural cues, while existing target-conditioned baselines remain limited by static or single-cue structural modeling. Compared with the target conditioning CRAFT, TCSD achieves clear improvements on unseen-dominant graphs, highlighting the importance of exploiting structural states and dynamics beyond target conditioning alone. Note that TAMI with DyGFormer fails on all unseen-dominant datasets due to out-of-memory issues.

EXP-2: Effectiveness of the proposed TCSD on seen-dominant datasets. TCSD achieves an average MRR improvement of 1.78% across eight seen-dominant datasets, with notable gains on mooc and uci of 11.97% and 3.77%, respectively. These results demonstrate the effectiveness of hybrid structural indicators and structural dynamics modeling. Compared with five target-conditioned baselines, TCSD achieves the best performance on 6 out of 8 datasets. Existing target-conditioned baselines are sensitive to structural regimes: co-neighbor-based methods perform well when shared-neighbor support is available, while repeat-based methods are competitive when recurrence dominates. For example, DyGFormer achieves second-best results on uci and Flights, where co-occurrence support is relatively high. In contrast, TCSD integrates exact and relaxed structural indicators and models their dynamics, enabling more consistent structural support modeling across regimes.

Across sixteen datasets, TCSD achieves larger gains on unseen-dominant graphs, further demonstrating the robustness of our structural dynamics modeling perspective.

5.3 Ablation Study

We conduct ablation studies on four datasets, including GoogleLocal, mooc, lastfm, and uci, to investigate the contribution of each component in TCSD. The results are reported in Table 3.

Table 3: Ablation study of TCSD on four datasets.

Variant	GoogleLocal	mooc	lastfm	uci
TCSD	66.24±0.55	69.78±0.38	55.88±0.45	79.50±0.19
w/o h_i^{re}	66.18±0.03	63.78±0.01	51.46±0.11	78.67±0.45
w/o $\widetilde{h}_i^{\mathrm{re}}$	50.64±0.25	61.11±0.14	47.74±0.70	76.27±0.10
w/o h_i^{co}	65.89±0.09	69.45±0.18	55.75±0.37	78.63±0.02
w/o $\widetilde{h}_i^{\mathrm{co}}$	63.95±0.34	64.25±0.12	51.39±0.28	78.56±0.07
w/o SSM	65.16±0.50	64.06±0.23	51.44±0.10	78.62±0.16
w/o SDA	62.10±2.64	59.19±2.62	51.32±0.25	78.14±0.25

EXP-3: Hybrid structural indicators provide complementary support. Removing any structural indicator degrades TCSD, confirming the necessity of hybrid structural indicators across diverse dynamic graphs, which is aligned with Fig. 3. Exact indicators are especially useful when explicit structural matches are available. Concretely, removing the exact repeat indicator h_i^{re} results in a relative MRR drop of 8.60% on mooc, while removing the exact co-neighbor indicator h_i^{co} leads to a relative drop of 1.09% on uci. This shows that repeat and co-neighbor matches provide high-confidence structural support when exact patterns are available. In contrast, such drops are smaller on those graphs (e.g., GoogleLocal and lastfm), where exact repeat and/or co-neighbor support are nearly absent. This aligns with our motivation that exact indicators preserve reliable signals when identity-based matches exist.

EXP-4: Relaxed indicators improve robustness in sparse-support regimes. Relaxed indicators $\widetilde{h}_i^{\mathrm{re}}$ and $\widetilde{h}_i^{\mathrm{co}}$ maintain informative support beyond exact identity matching. Removing either indicator consistently degrades performance across all four datasets. The largest degradation appears on GoogleLocal: removing $\widetilde{h}_i^{\mathrm{re}}$ reduces MRR by 23.55%, while removing $\widetilde{h}_i^{\mathrm{co}}$ reduces MRR by 3.46%. This confirms that when exact repeat or co-neighbor matches are sparse, relaxed embedding-based indicators can recover latent structural relevance that exact indicators fail to capture.

EXP-5: Effectiveness of structural dynamics modeling. When hybrid structural states are treated as static representations (w/o SSM), the performance shows an average relative MRR drop of 4.72% across the four datasets. The degradation is more evident on mooc and lastfm, with relative drops of 8.20% and 7.95%, respectively, where frequent repeated interactions make temporal consistency informative. This suggests that using structural indicators independently at each source-history position is insufficient; modeling how target-conditioned states evolve along source interactions helps distinguish consistent support from isolated cues, improving hard-negative disambiguation.

EXP-6: Effectiveness of active aggregation control. Removing structural-dynamics-guided aggregation (w/o SDA), where learned structural dynamics are used only as passive token features,

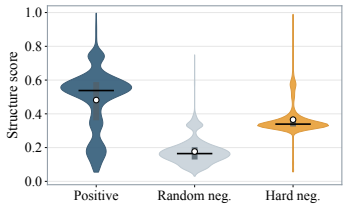


Figure 4: Normalized hybrid structural indicator score on uci.

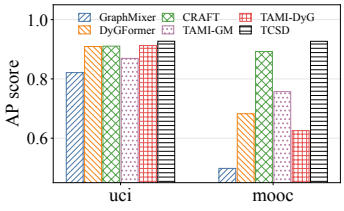


Figure 5: Hard-negative evaluation in AP on uci and mooc.

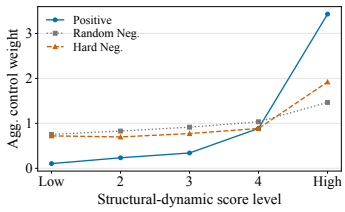


Figure 6: Visualization of aggregation contribution on mooc.

causes a 7.82% average relative MRR drop, with the largest drop on mooc (15.18%). This shows that structural dynamics are more useful when they actively guide aggregation rather than merely augment tokens. Without structural gates, these signals can be diluted by edge and time features, limiting the model's ability to emphasize target-relevant neighbors. These results support using structural states not only for encoding and evolution, but also as active aggregation guidance for link prediction.

5.4 Hard-Negative Disambiguation and Evaluation

We examine whether hybrid structural indicators can distinguish true links from hard negatives. For each positive test edge, we sample 100 negative destinations and select the hard negative with the highest exact structural score, computed from repeat interactions and co-neighbor overlap within recent histories (Fig. 1b). We compare the identity-based score with our normalized hybrid structural indicator score in Figs. 1b and 4.

EXP-7: Hybrid structural indicators reduce hard-negative ambiguity. In contrast to the hard-negative ambiguity caused by identity-based statistics in Fig. 1b, Fig. 4 shows that our hybrid indicators assign the highest scores to positive targets, averaging around 0.54, compared with 0.34 for hard negatives and 0.16 for random negatives. This indicates that combining exact and relaxed structural cues yields more discriminative target-conditioned structural states and suppresses misleading hard-negative evidence.

EXP-8: TCSD achieves stronger performance under hard-negative evaluation. We further evaluate whether hard-negative ambiguity affects link prediction. Using the same hard-negative construction protocol, we pair each positive test edge with its selected hard negative and compute AP (Average Precision) on uci and mooc. As shown in Fig. 5, TCSD achieves consistently higher AP scores on both datasets against five representative baselines, with up to 3.86% improvement. Several baselines are less robust: GraphMixer obtains only 0.498 AP on mooc, while DyGFormer and TAMI (DyGFormer) also degrade. Although CRAFT remains competitive, TCSD shows the most stable performance across datasets. These results show that TCSD improves not only indicator-level disambiguation but also prediction robustness against structurally plausible hard negatives.

5.5 Analysis on Structural Dynamics-Guided Aggregation

We examine whether TCSD aggregation is guided by target-conditioned structural dynamics. For each test query, we compute the hybrid structural score $h_i^{\mathrm{d}}(t)$ by averaging four normalized indicators and the normalized aggregation contribution $\chi_i(t) = |\mathcal{N}_s(t)|\alpha_i(t)$ for each valid historical neighbor, where $\alpha_i(t)$ is defined in Eq. (8). Thus, $\chi_i(t) > 1$ indicates a larger-than-uniform contribution. We rank all valid neighbor positions by $h_i^{\mathrm{d}}(t)$, split them into five equal-sized bins from Low to High, and report the mean $\chi_i(t)$ of each bin on the mooc dataset in Fig. 6.

EXP-9: Structural dynamics provide effective aggregation guidance. As shown in Fig. 6, TCSD adjusts neighbor contributions according to target-conditioned structural dynamics rather than aggregating them uniformly. Low-score neighbors are suppressed below the uniform baseline, while high-score neighbors are amplified. This is most evident for positive links on mooc, where the contribution increases from 0.11 to 3.43 across score bins, over a $31\times$ increase. Random and hard negatives show much weaker growth, suggesting that TCSD identifies target-relevant support rather than simply favoring active neighbors. These results support using structural indicators as active aggregation guidance, explaining the ablation trends in Table 3.

5.6 Effect of Signed Relaxed Indicators

To evaluate the effect of the sign of relaxed indicators, we construct a variant by normalizing the relaxed structural indicators into $[0, 1]$, denoted as TCSD-Norm. We report the results on two datasets in Table 4.

Table 4: Effect of signed relaxed structural indicators into $[0, 1]$ in MRR (%).

Dataset	TCSD	TCSD-Norm	Rel. change
uci	79.50	78.68	-1.03%
mooc	69.78	69.71	-0.10%

EXP-10: The signed relaxed indicators are useful. TCSD consistently outperforms TCSD-Norm, with TCSD-Norm showing relative MRR drops of 1.03% on the uci dataset and 0.10% on the mooc dataset. This suggests that the original signed cosine similarity provides useful directional contrast in structurally mixed regimes. Negative or weak similarities can help distinguish structurally plausible but less compatible neighbors from target-relevant ones. Since these signed indicators are further processed by structural dynamics and converted into gates, the convolutional transformations and MLPs can adaptively calibrate their scales and signs before aggregation.

5.7 Efficiency Evaluation

We compare the proposed TCSD against ten baselines in terms of per-epoch training time on mooc and lastfm, which contain $411{,}749$ and $1{,}293{,}103$ interactions, respectively.

EXP-11: TCSD achieves competitive time efficiency. As shown in Fig. 7, TCSD achieves competitive efficiency among the ten baselines, ranking as the third fastest method on both datasets. Although TCSD introduces target-conditioned structural state construction and dynamics-guided aggregation, it is substantially more efficient than most temporal and target-conditioned baselines. For example, TCSD is $4.43\times$ faster than DyGFormer on mooc and $3.76\times$ faster on lastfm; compared with CAWN, it is $5.31\times$ and $4.27\times$ faster, respectively. Compared with the target-conditioned baseline CRAFT, TCSD incurs moderate additional cost, taking $1.52\times$ and $1.59\times$ longer epoch time on mooc and lastfm, respectively. This overhead is reasonable given that TCSD explicitly models structural dynamics and achieves stronger prediction performance, indicating a practical trade-off between effectiveness and efficiency.

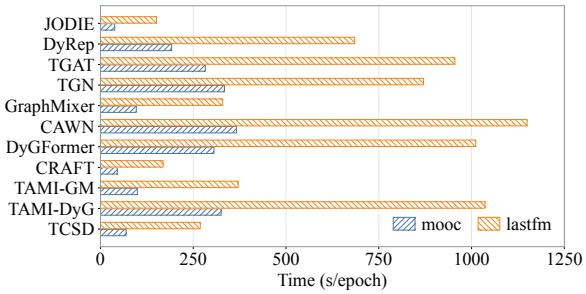


Figure 7: Per-epoch training time comparison on two datasets.

6 Conclusion

In this paper, we propose a new perspective on target-conditioned structural dynamics for link prediction in dynamic graphs. Based on this perspective, we propose TCSD, a target-conditioned structural dynamics framework. The key idea is to construct target-conditioned structural states as the basis, evolve these states to capture structural dynamics, and translate the learned dynamics into aggregation control, thereby moving structural signals from static passive feature injection to active aggregation guidance. TCSD designs hybrid structural indicators to construct target-conditioned structural states, combining exact indicators that preserve high-confidence structural support with relaxed indicators that mitigate encoding collapse under sparse structural regimes. We further propose a structural dynamics-guided aggregation mechanism that models the dynamics of these states and converts the learned dynamics into structural gates, allowing target-relevant historical interactions to be emphasized while irrelevant ones are suppressed. Extensive experiments on sixteen dynamic graphs show that TCSD outperforms ten competitive baselines, with larger gains on unseen-dominant datasets. Further analyses demonstrate that TCSD improves hard-negative disambiguation and verify the effectiveness of using structural dynamics as the active aggregation controller.

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The authors declare no competing interests.

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A Theoretical Analysis

A.1 Containment of Passive Structural Feature Injection

We show that a constant-coordinate extension of the proposed aggregation mechanism preserves the modeling capacity of a common passive structural feature injection design. The result is stated only at the aggregation stage: the context encoder is fixed, and the contextualized representations are

treated as aggregation inputs. Any destination representation used by the final predictor is shared by the compared aggregators and is omitted from the notation.

For a fixed query (s, d, t), we omit the explicit dependence on the timestamp t for notational simplicity. Let $\hat{\mathbf{z}}_i \in \widehat{\mathcal{Z}} \subset \mathbb{R}^D$ denote the contextualized non-structural representation of the i-th source-side historical interaction, and let $\mathbf{h}_i \in \mathcal{H} \subset \mathbb{R}^C$ denote its target-conditioned structural state. Assume $\widehat{\mathcal{Z}}$ and $\mathcal{H}$ are compact and the history length L is fixed. Let $\hat{\mathbf{Z}}_s = (\hat{\mathbf{z}}_1, \ldots, \hat{\mathbf{z}}_L)$ and $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_L)$. We consider passive structural feature injection of the following form:

$$F_{\mathrm{pc}}(\hat{\mathbf{Z}}_s, \mathbf{H}) = \rho_{\mathrm{pc}} \left(\sum_{i=1}^{L} \pi_i^{\mathrm{pc}} V_{\mathrm{pc}} \left([\hat{\mathbf{z}}_i \,\|\, \mathbf{h}_i] \right) \right), \tag{9}$$

where $\pi_i^{\mathrm{pc}} = \frac{\exp(a_{\mathrm{pc}}([\hat{\mathbf{z}}_i \,\|\, \mathbf{h}_i]))}{\sum_{j=1}^{L} \exp(a_{\mathrm{pc}}([\hat{\mathbf{z}}_j \,\|\, \mathbf{h}_j]))}$. Here, $a_{\mathrm{pc}} : \mathbb{R}^{D+C} \to \mathbb{R}$ is a continuous attention-logit function, $V_{\mathrm{pc}} : \mathbb{R}^{D+C} \to \mathbb{R}^P$ is an affine value projection, and $\rho_{\mathrm{pc}} : \mathbb{R}^P \to \mathbb{R}^Q$ is a continuous predictor. Throughout this subsection, $\|\cdot\|$ denotes the Euclidean norm on the corresponding finite-dimensional space. The identity value map is included as a special case. The affine restriction covers direct or linear passive injection before pooling, but does not cover nonlinear token-wise value maps inside the summation.

Proposition 1 (Aggregation-stage containment under constant-coordinate extension)**.** *At the aggregation stage, with the context encoder fixed, consider any passive structural injection aggregator F_{pc} of the form in Eq. (9). Let $\mathcal{F}_{\mathrm{TCSD}^+}$ denote the constant-coordinate extension of the TCSD aggregation family, obtained by replacing each contextualized representation with $\hat{\mathbf{z}}_i^+ = [2\hat{\mathbf{z}}_i \| \mathbf{1}_C] \in \mathbb{R}^{D+C}$. Suppose its gate network, attention-score network f_a, and predictor have sufficient width to uniformly approximate continuous functions on their corresponding compact domains. Then, for any such F_{pc} and any $\epsilon > 0$, there exists $F_{\mathrm{TCSD}^+} \in \mathcal{F}_{\mathrm{TCSD}^+}$ such that*

$$\sup_{(\hat{\mathbf{Z}}_s, \mathbf{H}) \in \widehat{\mathcal{Z}}^L \times \mathcal{H}^L} \left\| F_{\mathrm{TCSD}^+}(\hat{\mathbf{Z}}_s, \mathbf{H}) - F_{\mathrm{pc}}(\hat{\mathbf{Z}}_s, \mathbf{H}) \right\| < \epsilon. \tag{10}$$

Equivalently, if $\mathcal{F}_{\mathrm{pc}}$ denotes the family of aggregators of the form in Eq. (9), then $\mathcal{F}_{\mathrm{pc}} \subseteq \overline{\mathcal{F}_{\mathrm{TCSD}^+}}^{\|\cdot\|_\infty}$.

Proof. Since $\mathcal{H}$ is compact, each coordinate of $\mathbf{h} \in \mathcal{H}$ is bounded. Fix $\eta \in (0, 1/2)$ and choose a coordinate-wise affine normalization $N : \mathcal{H} \to [\eta, 1-\eta]^C$. For each non-constant coordinate, N rescales it into $[\eta, 1-\eta]$; for each constant coordinate, N maps it to $1/2$. Hence, on $N(\mathcal{H})$, there exists a coordinate-wise affine left inverse N^{-1} such that $N^{-1}(N(\mathbf{h})) = \mathbf{h}$ for all $\mathbf{h} \in \mathcal{H}$. On constant coordinates, this inverse is a constant map, which is a degenerate affine map.

Considering the augmented representation $\hat{\mathbf{z}}_i^+ = [2\hat{\mathbf{z}}_i \,\|\, \mathbf{1}_C] \in \mathbb{R}^{D+C}$, the structural-dynamics output can be ignored by the gate network in this construction. Let the structural gate approximate

$$\mathbf{g}_i^+ = [\tfrac{1}{2}\mathbf{1}_D \,\|\, N(\mathbf{h}_i)]. \tag{11}$$

The first D coordinates have logit target zero, yielding $1/2$ after the sigmoid when exactly attained. The last C coordinates have logit target $\mathrm{logit}(N(\mathbf{h}_i))$, which is continuous on $\mathcal{H}$ since $N(\mathbf{h}) \in [\eta, 1-\eta]^C \subset (0,1)^C$. A sufficiently wide gate MLP can uniformly approximate this joint logit target.

Thus, for any $\varepsilon_g > 0$, the gated representation $\widetilde{\mathbf{z}}_i^+ = \mathbf{g}_i^+ \odot \hat{\mathbf{z}}_i^+$ can be made to satisfy

$$\max_i \left\| \widetilde{\mathbf{z}}_i^+ - \boldsymbol{\xi}_i \right\| < \varepsilon_g, \;\; \boldsymbol{\xi}_i := [\hat{\mathbf{z}}_i \,\|\, N(\mathbf{h}_i)], \tag{12}$$

uniformly over $\widehat{\mathcal{Z}}^L \times \mathcal{H}^L$.

Let $\delta > 0$ and define the compact neighborhood $\mathcal{K}_\delta = \{[\hat{\mathbf{z}} \| \boldsymbol{\omega}] : \mathrm{dist}(\hat{\mathbf{z}}, \widehat{\mathcal{Z}}) \le \delta, \boldsymbol{\omega} \in [\eta/2, 1 - \eta/2]^C\}$. Extend N^{-1} affinely to $[\eta/2, 1-\eta/2]^C$, using a constant extension on constant coordinates, and define

$$\bar{a}([\hat{\mathbf{z}} \,\|\, \boldsymbol{\omega}]) := a_{\mathrm{pc}}([\hat{\mathbf{z}} \,\|\, N^{-1}(\boldsymbol{\omega})]) \tag{13}$$

on $\mathcal{K}_\delta$. Then $\bar{a}$ is continuous on $\mathcal{K}_\delta$ and $\bar{a}(\boldsymbol{\xi}_i) = a_{\mathrm{pc}}([\hat{\mathbf{z}}_i \| \mathbf{h}_i])$. By choosing ε_g sufficiently small, all $\widetilde{\mathbf{z}}_i^+$ lie in $\mathcal{K}_\delta$. A sufficiently wide attention-logit network f_a can uniformly approximate $\bar{a}$ on $\mathcal{K}_\delta$, and the TCSD attention logits can therefore be made uniformly close to the passive logits.

Since the softmax map is uniformly continuous on bounded domains and L is fixed, the resulting TCSD attention weights α_i can be made uniformly close to π_i^{pc}. That is, for any $\varepsilon_\alpha > 0$,

$$\|\boldsymbol{\alpha} - \boldsymbol{\pi}^{\mathrm{pc}}\|_1 < \varepsilon_\alpha \tag{14}$$

uniformly over $\widehat{\mathcal{Z}}^L \times \mathcal{H}^L$.

Because V_{pc} and N^{-1} are affine, there exists an affine map $\Gamma_{\mathrm{pc}} : \mathbb{R}^{D+C} \to \mathbb{R}^P$ such that

$$\Gamma_{\mathrm{pc}}\left([\hat{\mathbf{z}} \,\|\, N(\mathbf{h})]\right) = V_{\mathrm{pc}}\left([\hat{\mathbf{z}} \,\|\, \mathbf{h}]\right), \ \forall (\hat{\mathbf{z}}, \mathbf{h}) \in \widehat{\mathcal{Z}} \times \mathcal{H}. \tag{15}$$

Let $\boldsymbol{\mu}_{\mathrm{TCSD}^+} = \sum_{i=1}^L \alpha_i \widetilde{\mathbf{z}}_i^+$ and $\boldsymbol{\mu}_{\mathrm{pc}} = \sum_{i=1}^L \pi_i^{\mathrm{pc}} V_{\mathrm{pc}}\left([\hat{\mathbf{z}}_i \,\|\, \mathbf{h}_i]\right)$. Since Γ_{pc} is affine and the attention weights sum to one, Eq. (15) gives

$$\|\Gamma_{\mathrm{pc}}(\boldsymbol{\mu}_{\mathrm{TCSD}^+}) - \boldsymbol{\mu}_{\mathrm{pc}}\| \le \|\Gamma_{\mathrm{pc}}\|_{\mathrm{Lip}} \max_i \left\|\widetilde{\mathbf{z}}_i^+ - \boldsymbol{\xi}_i\right\| + M_\Gamma \|\boldsymbol{\alpha} - \boldsymbol{\pi}^{\mathrm{pc}}\|_1, \tag{16}$$

where

$$M_\Gamma = \sup_{(\hat{\mathbf{Z}}_s, \mathbf{H}) \in \widehat{\mathcal{Z}}^L \times \mathcal{H}^L} \max_i \|\Gamma_{\mathrm{pc}}(\boldsymbol{\xi}_i)\| < \infty$$

by compactness. The two terms are controlled by Eqs. (12) and (14), respectively. Thus, $\Gamma_{\mathrm{pc}}(\boldsymbol{\mu}_{\mathrm{TCSD}^+})$ can be made uniformly close to $\boldsymbol{\mu}_{\mathrm{pc}}$.

Finally, let the TCSD predictor approximate the continuous map $\boldsymbol{\mu} \mapsto \rho_{\mathrm{pc}}(\Gamma_{\mathrm{pc}}(\boldsymbol{\mu}))$ on the compact image of the TCSD pooling map. Since the relevant inputs form a compact set and ρ_{pc} is continuous, choosing the gate-copy, attention-logit, and predictor approximation errors sufficiently small yields

$$\sup_{(\hat{\mathbf{Z}}_s, \mathbf{H}) \in \widehat{\mathcal{Z}}^L \times \mathcal{H}^L} \left\| F_{\mathrm{TCSD}^+}(\hat{\mathbf{Z}}_s, \mathbf{H}) - F_{\mathrm{pc}}(\hat{\mathbf{Z}}_s, \mathbf{H}) \right\| < \epsilon. \tag{17}$$

Therefore, $\mathcal{F}_{\mathrm{pc}} \subseteq \overline{\mathcal{F}_{\mathrm{TCSD}^+}}^{\|\cdot\|_\infty}$. □

> **Remark.** Proposition 1 shows that the constant-coordinate extension of TCSD, denoted TCSD$^+$, can recover aggregation-stage passive structural injection with identity or affine value projection up to arbitrary uniform approximation. In this sense, passive structural injection is subsumed by TCSD$^+$ rather than being an additional mechanism outside our aggregation framework.

A.2 Time Complexity Analysis

We analyze the mini-batch time complexity of TCSD. Let B denote the batch size, q the number of candidate destinations per interaction, L the sampled neighbor count, D the hidden dimension, d_x the embedding dimension used for relaxed similarities, and N_Φ the number of context-encoder layers. For each candidate link, source/destination context encoding costs $O(N_\Phi(L^2D + LD^2))$. Constructing hybrid structural states costs $O(L^2 d_x)$. The structural dynamics and gating modules operate on the length-L source history and incur at most $O(LD^2)$ cost, which is absorbed by the encoder term when $N_\Phi \ge 1$. Thus, the mini-batch complexity is $O(Bq(N_\Phi(L^2D + LD^2) + L^2 d_x))$. When $d_x = O(D)$ and N_Φ is fixed, this simplifies to $O(Bq(L^2D + LD^2))$.

B Full Structural Support Statistics

To complement the representative examples in Fig. 1a, we report structural support statistics for all sixteen datasets in Fig. 8. We compute two statistics on test positive links. The repeat ratio measures the fraction of links whose destination node appears in the source node's historical neighborhood. The co-occurrence support ratio measures the fraction of links whose source-side history overlaps with the destination-side history, where co-neighbor overlap is computed using 20 recently sampled neighbors.

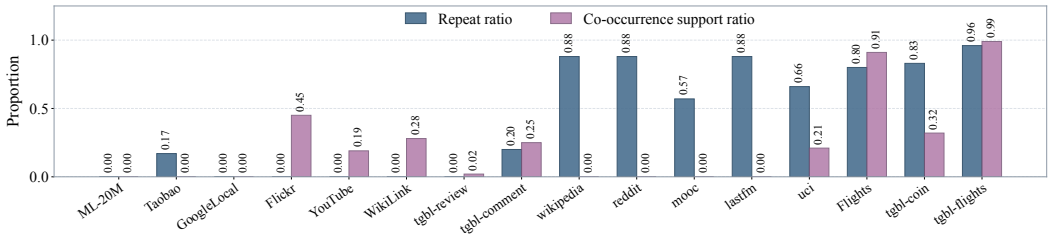


Figure 8: Structural support statistics of the test set over sixteen datasets. The blue bars denote the repeat ratio, and the purple bars denote the co-occurrence support ratio.

As shown in Fig. 8, dynamic graphs exhibit substantially different structural regimes. Some datasets contain strong repeat evidence, while others provide little or no exact repeat support. Similarly, co-occurrence support is highly dataset-dependent and appears only in certain graphs. These observations further confirm that relying on a single identity-based structural cue is insufficient for robust dynamic link prediction. They also support the design of TCSD, which jointly models repeat-based and co-neighbor-based evidence with both exact and relaxed structural indicators.

C Additional Experiments

C.1 Detailed Experimental Settings

Datasets. Table 5 provides detailed descriptions of the sixteen datasets, including node count, edge count, timestamps, repeat ratio, and domains. These datasets are collected from [22, 59, 60, 61, 62, 63]. Following the standard protocol [59, 64], we chronologically split each dataset into training, validation, and test sets with a ratio of 70%, 15%, and 15%.

Implementation details. For TCSD, we use recent-neighbor sampling with 20 neighbors and train the model with a batch size of 256, a learning rate of 3×10^{-4}, and cross-entropy loss. The hidden dimension is set to 172, the encoder contains 2 layers, and the convolution operation contains 1 layer. We set the input dropout to 0.1 and use hidden/attention dropout of 0.1 for most datasets; for sparser datasets, we increase the hidden and attention dropout to 0.5 when selected by validation. All experiments were conducted on a Linux server with 8 Tesla V100-SXM2 GPUs, each with 32GB memory, and two Intel Xeon Gold 6240 CPUs with 376 GB of system memory. For baselines marked with $*$, we use the reported results from CRAFT [22]. For baselines without $*$, we rerun the official implementation using our fixed splits and negative samples.

To ensure a fair comparison, all methods, including TCSD and all baselines such as CRAFT, are implemented and evaluated within the DyGLib [19]. We use identical data splits, negative samples, evaluation protocols, and early-stopping criteria across all methods.

Baselines. We compare TCSD with ten representative and the latest dynamic graph link prediction methods. In particular, we include CRAFT [22] and TAMI [18], two recent works that are tailored for dynamic link prediction. CRAFT represents the repeat-aware line of work, as it explicitly models recurring interactions for future link prediction and serves as a strong baseline on repeat-dominant dynamic graphs. TAMI is a recent framework for handling heterogeneous temporal interactions through log time encoding and link history aggregation. Although TAMI is not a dedicated target-conditioned structural encoding method, it can be instantiated with both general temporal graph backbones and co-neighbor-based target-conditioned backbones, such as GraphMixer and DyGFormer. Including TAMI, therefore, allows us to compare TCSD with a heterogeneity-aware framework that improves existing temporal link prediction models. We also include DyGFormer, a widely used and competitive co-neighbor-based dynamic graph model that captures target-dependent structural support through neighborhood overlap. Together, these baselines cover repeat-aware modeling, heterogeneity-aware temporal interaction modeling, and co-neighbor-based target-conditioned structural modeling.

Table 5: Dataset statistics

Dataset	Nodes (users/items)	Edges	Timestamps	Repeat ratio(%)	Bipartite	Domain
ML-20M	100,785/9,646	14,494,325	9,993,250	0	✓	Movie rating
Taobao	760,617/863,016	18,853,792	139,171	16.58	✓	E-commerce interaction
GoogleLocal	206,244/267,336	1,913,967	1,771,060	0	✓	Business review
Flickr	233,836	7,223,559	134	0	×	Who-To-Follow
YouTube	402,422	3,288,028	203	0	×	Who-To-Follow
WikiLink	1,361,972	34,163,774	2,198	0	×	Web link
tgbl-review	352,636/298,590	4,873,540	6,865	0.19	✓	E-commerce review
tgbl-coin	638,486	22,809,486	1,295,720	82.93	×	Transaction
tgbl-comment	994,790	44,314,507	30,998,030	19.81	×	Reply network
tgbl-flights	18,143	67,169,570	1,385	96.48	×	Transport
wikipedia	8,227/1,000	157,474	152,757	88.41	✓	Interaction
reddit	10,000/984	672,447	669,065	88.32	✓	Reply network
mooc	7,047/97	411,749	345,600	56.66	✓	Interaction
lastfm	980/1,000	1,293,103	1,283,614	88.01	✓	Interaction
uci	1,899	59,835	58,911	66.06	×	Social contact
Flights	13,169	1,927,145	122	79.50	×	Transport

Table 6: Transductive dynamic link prediction comparison on four datasets under historical and inductive negative samplings. The **best** and second-best results are highlighted accordingly.

Historical Sampling								
Model	AP (%)				AUC (%)			
	uci	mooc	wikipedia	reddit	uci	mooc	wikipedia	reddit
GraphMixer	85.00±0.15	44.10±0.02	91.10±0.02	77.38±0.07	79.00±0.21	39.10±0.02	88.06±0.10	77.34±0.05
DyGFormer	78.21±0.11	77.74±0.07	80.54±0.02	81.71±0.00	72.68±0.11	73.04±0.01	76.15±0.02	81.02±0.02
CRAFT	91.37±0.02	95.31±0.03	97.60±0.00	97.04±0.03	87.54±0.03	95.69±0.05	96.82±0.01	97.07±0.01
TAMI-GM	63.03±0.31	84.08±0.05	69.10±0.09	73.22±0.13	53.84±0.14	85.60±0.04	67.11±0.08	72.63±0.07
TAMI-DyG	75.35±0.30	82.37±0.01	78.95±0.25	81.27±0.03	70.35±0.17	81.79±0.02	74.60±0.03	80.19±0.04
TCSD	**94.29±0.10**	**96.13±0.01**	**98.65±0.02**	**97.40±0.01**	**92.03±0.14**	**96.57±0.04**	**98.39±0.00**	**97.62±0.01**
Rel. Imprv.	+3.20%	+0.86%	+1.08%	+0.37%	+5.13%	+0.92%	+1.62%	+0.57%
Inductive Sampling								
Model	AP (%)				AUC (%)			
	uci	mooc	wikipedia	reddit	uci	mooc	wikipedia	reddit
GraphMixer	83.25±0.03	49.59±0.05	89.62±0.03	85.95±0.06	78.01±0.12	49.24±0.05	85.59±0.01	83.38±0.01
DyGFormer	71.31±0.41	78.06±0.06	79.14±0.05	91.47±0.00	64.36±0.30	73.82±0.04	73.03±0.08	86.84±0.00
CRAFT	91.47±0.03	93.35±0.08	97.51±0.06	97.89±0.06	88.35±0.03	94.48±0.05	96.84±0.04	97.64±0.04
TAMI-GM	62.02±0.56	77.61±0.18	62.52±0.07	79.84±0.06	54.35±0.49	79.79±0.15	59.28±0.04	75.65±0.04
TAMI-DyG	66.47±0.32	77.98±0.01	80.20±0.01	89.97±0.00	60.52±0.44	74.78±0.00	74.71±0.02	85.03±0.00
TCSD	**93.23±0.13**	**95.36±0.11**	**98.41±0.03**	**98.20±0.07**	**91.00±0.03**	**95.94±0.04**	**98.18±0.02**	**98.08±0.04**
Rel. Imprv.	+1.92%	+2.15%	+0.92%	+0.32%	+3.00%	+1.55%	+1.38%	+0.45%

C.2 Effectiveness Evaluation under Historical and Inductive Negative Samplings

We further evaluate TCSD on four transductive link prediction datasets using AP and AUC, comparing against recent representative baselines. Here, we use two temporal sampling schemes: historical negative sampling and inductive negative sampling. We report these results of our TCSD and recent representative baselines in Table 6.

EXP-12: TCSD better disambiguates historical negatives. Under historical negative sampling, negative candidates are more likely to have appeared in the historical context, making them structurally plausible and harder to distinguish from positives. As shown in Table 6, TCSD consistently achieves the best AP and AUC across all four datasets, with up to 3.20% AP and 5.13% AUC improvements on uci. This demonstrates that TCSD can better identify reliable target-relevant structural support and suppress misleading historical evidence, validating the benefit of hybrid structural states and dynamics-guided aggregation for hard-negative disambiguation.

EXP-13: TCSD generalizes well under the inductive negative sampling. Under inductive negative sampling, candidate negatives are less exposed in the historical training context, requiring the model to generalize beyond memorized repeat patterns. It is observed from Table 6 that TCSD remains consistently superior, achieving up to 2.15% AP improvement on mooc and 3.00% AUC improvement on uci. These results show that TCSD is not merely exploiting historical recurrence; by combining

exact and relaxed structural indicators and modeling their dynamics, it captures more generalizable structural support for future link prediction under evolving graph structures.

C.3 Indicator Complementarity across Structural Regimes

We first examine whether different structural indicators provide complementary signals across graph regimes. For each test query, we compute the four structural indicators over the recently sampled neighbors for both the positive target and sampled negative targets. For each indicator dimension c, we use the separability gap $\Delta_c = \mathbb{E}[\bar{H}_c^+] - \mathbb{E}[\bar{H}_c^-]$, where $\bar{H}_c^+$ and $\bar{H}_c^-$ denote the average indicator scores over positive and negative targets, respectively. A larger Δ_c indicates that the indicator better distinguishes true future neighbors from random candidates. Fig. 3 reports the indicator-wise separability across six datasets, whose structural regimes are summarized in Appendix B.

EXP-14: Structural indicators are complementary across graph regimes. We observe that no single indicator dominates across all datasets, confirming that dynamic graphs exhibit heterogeneous structural regimes. Repeat-oriented indicators are more discriminative on repeat-driven datasets such as uci, mooc, and wikipedia; for example, the relaxed repeat indicator shows the largest separability gaps on wikipedia (0.754), and the exact repeat indicator is also strong on uci (0.121) and wikipedia (0.356). In contrast, co-neighbor indicators become more informative on structurally rich datasets such as Flights (0.343), where shared neighbors provide useful structural evidence.

EXP-15: Relaxed indicators offer more stable structural separability. Relaxed indicators show positive separability on almost all datasets and are less affected by the absence of exact identity matches. The relaxed repeat indicator is consistently strong, with gaps from 0.166 to 0.754 across the six datasets, whereas the exact repeat indicator drops to 0.00 on YouTube and GoogleLocal. A similar pattern appears for co-neighbor evidence: the relaxed co-neighbor indicator remains informative on Flights, YouTube, and GoogleLocal, while the exact co-neighbor indicator is close to zero or unavailable on several datasets. These results suggest that exact indicators are useful when exact repeat or co-neighbor matches exist, but they easily become uninformative under sparse support. In contrast, relaxed indicators provide a relaxed structural signal that captures latent structural patterns beyond exact identity-based matching, which is essential for handling diverse real-world dynamic structural regimes.

C.4 Parameter Sensitivity Analysis

We evaluate the sensitivity of TCSD to the number of sampled historical neighbors on two datasets. We report their results in Fig. 9.

EXP-16: Robustness towards the hyperparameter of TCSD. We observe that TCSD is relatively robust when the number of sampled neighbors ranges from 15 to 25. As the number of neighbors increases from 1 to 15, the MRR improves rapidly on both datasets, suggesting that richer recent histories provide more reliable target-conditioned structural dynamics. However, using too many neighbors may introduce stale or noisy interactions and increase computational cost. Therefore, we set the number of sampled historical neighbors to 20 by default.

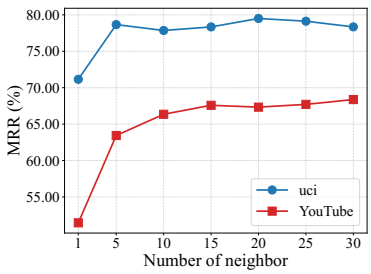


Figure 9: Effect of the number of neighbors on two datasets.

C.5 Effect of Passive Structural Feature Concatenation

We conduct a variant study on mooc to quantify how passive concatenation affects the use of structural information during aggregation. We compare TCSD with *w/o SDA*, which retains the same four structural indicators and structural dynamics modeling module but replaces the structural controller with passive concatenation of the learned dynamics with time and edge features. The analysis covers $61,546$ test queries with nonempty histories and $1,185,042$ valid historical-neighbor positions. We partition all valid neighbor positions into five quantile-based groups, Q_1–Q_5, according to the hybrid structural score $h_i^d(t)$, keeping tied scores together. Group sizes may therefore differ. For each group, we report the mean normalized aggregation contribution $\chi_i(t) = |\mathcal{N}_s(t)|\alpha_i(t)$, where $|\mathcal{N}_s(t)|$ counts valid sampled historical neighbors and $\alpha_i(t)$ is the pooling weight normalized over these positions.

Thus, $\chi_i(t) = 1$ corresponds to uniform aggregation. The contribution contrast Q_5/Q_1 is computed as the ratio of the unrounded mean contributions in the highest- and lowest-score groups. We report these results in Table 7.

EXP-17: Active aggregation control preserves stronger structural discrimination than passive concatenation. TCSD assigns a normalized contribution of 0.14 to Q_1 and 3.37 to Q_5, suppressing structurally less relevant neighbors while emphasizing more relevant ones. In contrast, these values become 0.39 and 2.34 without SDA, indicating weaker suppression of low-relevance neighbors and weaker emphasis on high-relevance neighbors. Consequently, the contribution contrast between Q_5 and Q_1 decreases from 24.07 to 6.00, a relative reduction of 75.1%. This is accompanied by a relative MRR drop of 15.18%, from 69.78 to 59.19. These results show that structural dynamics are less effective at emphasizing target-relevant neighbors and suppressing irrelevant ones when used only as passive token features. Using them to actively guide aggregation better adjusts neighbor contributions according to structural relevance and improves link prediction. This aligns with our motivation to use structural dynamics as active aggregation controllers rather than passive token features.

Table 7: Aggregation contribution of TCSD and passive concatenation on mooc. Values are rounded; contrasts use unrounded means.

Variant	MRR	Q_1	Q_2	Q_3	Q_4	Q_5	Q_5/Q_1
w/o SDA	59.19	0.39	0.48	0.64	1.19	2.34	6.00
TCSD	**69.78**	0.14	0.25	0.37	0.88	3.37	**24.07**

C.6 Clarification

While TCSD shows strong performance under multiple experimental settings, this work focuses on link prediction. Extending target-conditioned structural dynamics to graph-level or subgraph-level prediction [65, 59] requires task-specific definitions of structural states, which we leave for future work.

C.7 Broader Impacts

Dynamic graphs appear in many real-world systems [66, 23], including recommender systems, social networks, traffic systems, and multi-agent systems. By modeling target-conditioned structural dynamics, TCSD may improve prediction robustness, which can enhance the practical utility of dynamic graph learning. We do not identify any direct negative societal impacts of this work.

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